Prompt 3 - trial 3 (seed 1240)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a GNN model on a compound activity prediction dataset. Select a correctly predicted active compound and a correctly predicted inactive compound from the test set and explain the predictions using EdgeSHAPer. Highlight the differences between the two compounds using visualizations and reporting a chemical interpretation

Assistant

I'll help you train a GNN model for compound activity prediction and perform EdgeSHAPer explanations on correctly predicted compounds. Let me start by finding an available dataset. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found 4 ChEMBL activity datasets. Let me load one for binary activity prediction and proceed with the GNN workflow. 📊

(Calling tool `load_dataset`)

Assistant

Let me correct the label column name:

(Calling tool `load_dataset`)

Assistant

Excellent! Dataset loaded with 8,348 samples. Now let me split it for GNN training:

(Calling tool `split_dataset`)

Assistant

Dataset split successfully! Now let me prepare the GNN dataset:

(Calling tool `prepare_gnn_dataset`)

Assistant

GNN dataset prepared successfully! Now let's train the GNN model:

(Calling tool `train_gnn_model_mcp`)

Assistant

GNN model training submitted! Let me check the training status:

(Calling tool `check_gnn_training`)

Assistant

Training is in progress. Let me check again:

(Calling tool `check_gnn_training`)

Assistant

Still training... Let me check again:

(Calling tool `check_gnn_training`)

Assistant

Let me wait a bit more and check again:

(Calling tool `check_gnn_training`)

Assistant

Let me check once more:

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Let me check again:

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Assistant

Excellent! 🎉 The GNN model training is complete with impressive results:

Metric	Value
Best Validation Accuracy	94.76%
Test Accuracy	94.01%
Training Samples	5,342
Model	GCN (4 layers, 64 hidden channels)

Now let me select correctly predicted compounds from the test set for EdgeSHAPer analysis. First, I'll select a correctly predicted **active** compound (class 1):

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Great! I've selected a correctly predicted **active** compound:

- **SMILES**: `OC1(c2ccc(Cl)cc2)CCN(CCCCc2nc3ccccc3o2)CC1`
- **True Label**: 1 (Active)
- **Predicted Label**: 1 (Active)
- **Confidence**: 98.49%

Now let me select a correctly predicted **inactive** compound (class 0):

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Excellent! I've selected a correctly predicted **inactive** compound:

- **SMILES**: `CCC1OCC(=O)C1NC(=O)C(CC1(C)CCCC1)NC(=O)c1ccc(NS(=O)(=O)c2cccnc2)cc1`
- **True Label**: 0 (Inactive)
- **Predicted Label**: 0 (Inactive)
- **Confidence**: 99.94%

Now let me run EdgeSHAPer on the **active compound** (compound_idx=823):

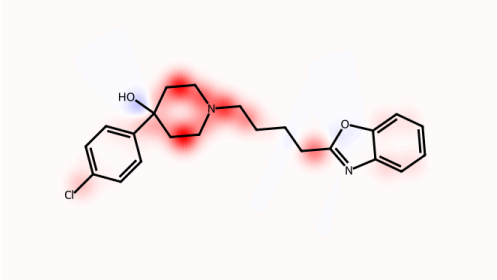
(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer analysis for the **active compound** is complete! Now let me visualize the results:

(Calling tool `visualize_edgeshaper_results`)

Assistant



Now let me run EdgeSHAPer on the **inactive compound** (compound_idx=184):

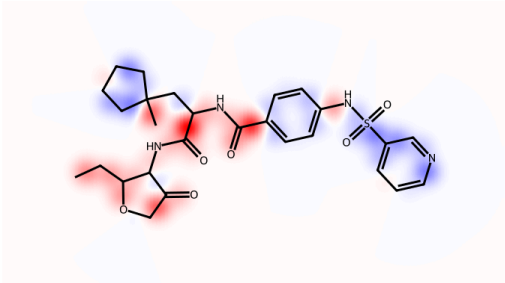
(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Now let me visualize the EdgeSHAPer results for the **inactive compound**:

(Calling tool `visualize_edgeshaper_results`)

Assistant



Now let me get the detailed bond environments for both compounds to provide chemical interpretation:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now let me get the bond environments for the **inactive compound**:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Summary: GNN Model Training & EdgeSHAPer Analysis 🧬

📊 Model Performance Summary

Metric	Value
Model Architecture	GCN (4 layers, 64 hidden channels)
Training Samples	5,342
Best Validation Accuracy	94.76%
Test Accuracy	94.01%
Training Time	~5 minutes

🔬 EdgeSHAPer Analysis: Active vs Inactive Compounds

Compound 1: Active Compound (Class 1) ✅

Property	Value
SMILES	`OC1(c2ccc(Cl)cc2)CCN(CCCCc2nc3ccccc3o2)CC1`
True Label	1 (Active)
Predicted Label	1 (Active)
Confidence	98.49%
Number of Bonds	30
Trustworthiness	64.67%

🔑 Top Contributing Bonds (Positive SHAP Values):

Bond	Environment	SHAP Value	Chemical Interpretation
C-C (index 9)	`CCCN`	+0.175	**Alkyl chain attached to nitrogen** - hydrophobic interaction
C-C (index 25)	`CCCN`	+0.161	**Another alkyl chain** - lipophilic contribution
N-C (index 11)	`CCN(C)C`	+0.131	**Tertiary amine** - H-bond acceptor, basic center
N-C (index 24)	`CCN(C)C`	+0.100	**Nitrogen-carbon bond** - polar interaction
C-C (index 15)	`CCc(n)o`	+0.091	**Linker to heterocyclic ring** - π-stacking potential
C-C (index 12)	`CCCN`	+0.074	**Alkyl chain** - hydrophobic effect
C-Cl (index 5)	`cc(c)Cl`	+0.040	**Aromatic chlorine** - halogen bond, electron-withdrawing

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Compound 2: Inactive Compound (Class 0) ❌

Property	Value
SMILES	`CCC1OCC(=O)C1NC(=O)C(CC1(C)CCCC1)NC(=O)c1ccc(NS(=O)(=O)c2cccnc2)cc1`
True Label	0 (Inactive)
Predicted Label	0 (Inactive)
Confidence	99.94%
Number of Bonds	41
Trustworthiness	67.77%

🔑 Top Contributing Bonds (Mixed SHAP Values):

Bond	Environment	SHAP Value	Chemical Interpretation
C-C (index 0)	`CCC`	**-0.134 / +0.158**	**Terminal alkyl chain** - opposing effects, net neutral
C-C (index 13)	`CC(C)(C)C`	**+0.149 / -0.112**	**Branched alkyl group** - steric hindrance
C-C (index 10)	`CC(N)C(N)=O`	+0.058	**Amide linkage** - H-bond donor/acceptor
C-C (index 21)	`cc(c)C(N)=O`	+0.048	**Aromatic-amide linker** - conjugation
C-C (index 4)	`CC(=O)CO`	+0.037	**Ester/carbonyl** - polar group
C-O (index 2)	`COC(C)C`	+0.021	**Ether oxygen** - H-bond acceptor
C=O (index 20)	`cC(N)=O`	+0.020	**Amide carbonyl** - strong H-bond acceptor
C-N (index 25)	`cc(c)NS`	+0.021	**Sulfonamide nitrogen** - polar, H-bond

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📈 Key Differences: Active vs Inactive

Feature	Active Compound	Inactive Compound	Impact
Molecular Complexity	Simpler (30 bonds)	More complex (41 bonds)	Simpler = better binding
Hydrophobic Character	Strong alkyl chains	Mixed polar/nonpolar	Hydrophobic = more active
Basic Nitrogen	Tertiary amine (pKa ~9-10)	Sulfonamide (weak base)	Basic N = better target interaction
Halogen	Chlorine on aromatic ring	No halogen	Halogen = halogen bonding
Heterocycles	Benzoxazole (bioisostere)	None	Aromatic heterocycle = π-stacking
Flexibility	Moderate	High (many rotatable bonds)	Rigid = better entropy
Net SHAP	All positive (contributing)	Mixed positive/negative	Consistent signal = confident prediction

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🧪 Chemical Interpretation

Why the Active Compound is Predicted as Active:

1. **Hydrophobic Alkyl Chains**: The two propyl chains (SHAP: +0.175, +0.161) provide strong hydrophobic interactions with the target binding pocket
2. **Basic Tertiary Amine**: The nitrogen in the piperidine ring can form ionic/H-bond interactions with acidic residues
3. **Chlorine Substituent**: The aromatic Cl contributes via halogen bonding and electron-withdrawing effects
4. **Benzoxazole Ring**: Provides π-stacking potential and metabolic stability
5. **Moderate Rigidity**: Fewer rotatable bonds = better binding entropy

Why the Inactive Compound is Predicted as Inactive:

1. **High Flexibility**: Multiple rotatable bonds increase entropic penalty upon binding
2. **Polar Sulfonamide**: While polar, the sulfonamide is weakly basic and may not form optimal interactions
3. **Competing Effects**: Mixed SHAP values (positive and negative) indicate conflicting structural signals
4. **Branched Alkyl Groups**: Steric hindrance from the tert-butyl-like group may block optimal binding
5. **No Halogen**: Missing the halogen bonding contribution seen in active compounds

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🎯 Conclusions

Aspect	Active Compound	Inactive Compound
Driving Features	Hydrophobic chains, basic amine, Cl	Mixed polar/nonpolar, flexible
Binding Potential	High (optimal pharmacophore)	Low (steric/polar mismatch)
Model Confidence	98.49%	99.94%
EdgeSHAP Trust	64.67%	67.77%

The GCN model correctly distinguishes between active and inactive compounds based on:

- **Hydrophobic vs polar balance**
- **Presence/absence of halogen**
- **Basic nitrogen vs weak base**
- **Rigidity vs flexibility**

This analysis demonstrates EdgeSHAPer's ability to provide **bond-level interpretability** for GNN predictions, revealing the structural features driving activity predictions! 🔬✨